

Isreal Cheruiyot

Aspiring RISC-V / CPU Design Engineer

Washington, USA | Email | Phone | GitHub | LinkedIn

SUMMARY

Aspiring computer engineer focused on **RISC-V processor architecture, CPU design, digital logic, FPGA development, and low-level programming**. Experienced with digital logic concepts, ALU and full-adder design, computer architecture, assembly language, and C/C++. Interested in designing and implementing efficient processors and hardware systems from the logic level through RTL and FPGA implementations.

TECHNICAL SKILLS

Computer Architecture

- RISC-V ISA and instruction-set concepts
- CPU datapath and control logic
- ALU design
- Registers, multiplexers, decoders, and memory systems
- Pipelining and processor architecture
- Digital logic and Boolean algebra

Hardware Design

- RTL design
- Verilog / SystemVerilog
- FPGA development
- Combinational and sequential logic
- Hardware protocols
- Electrical schematics and circuit design
- Digital circuit simulation

Programming

- C / C++

- Assembly Language
- Python
- HTML / CSS
- Git / GitHub

Tools

- VS Code
- Git / GitHub
- QEMU
- FPGA development tools
- Logic simulation tools

PROJECTS

RISC-V CPU / Processor Project

Personal Project

- Designed and implemented components of a RISC-V-inspired processor architecture.
- Developed an understanding of instruction fetch, decode, execute, memory, and write-back stages.
- Implemented fundamental CPU components including an ALU, registers, multiplexers, and control logic.
- Studied RISC-V instruction formats and assembly-level execution.
- Used simulation to verify processor behavior and troubleshoot hardware logic.

Digital Logic & ALU Design

Personal / Academic Project

- Designed digital logic circuits using fundamental gates and Boolean logic.
- Built and analyzed full adders and arithmetic circuits.
- Designed an ALU capable of performing arithmetic and logical operations.

- Applied concepts including multiplexing, binary arithmetic, combinational logic, and sequential logic.
- Connected individual logic components to understand how larger CPU subsystems are constructed.

Assembly & Low-Level Programming

Personal Project

- Wrote and analyzed assembly programs to understand how high-level instructions translate into machine operations.
- Studied registers, memory addressing, instruction encoding, branching, and arithmetic operations.
- Explored the relationship between assembly instructions and CPU datapath operations.
- Used emulation/simulation environments to experiment with low-level processor behavior.

EDUCATION

High School

Washington, USA

Expected Graduation: 20XX

Relevant Coursework

- Algebra II Honors
- Computer Science
- Digital Logic
- Computer Architecture
- Programming
- Physics

RELEVANT EXPERIENCE

Independent Computer Engineering Study

Self-Directed

- Studying CPU architecture, RISC-V, assembly language, digital logic, and hardware design.
- Building progressively more complex computer systems from individual logic components.
- Exploring the relationship between instruction-set architecture, RTL, FPGA implementations, and physical processors.
- Developing projects to strengthen hardware and low-level programming skills.

INTERESTS

- RISC-V Processor Architecture
- CPU / Microarchitecture
- FPGA Engineering
- ASIC Design
- RTL Design
- Digital Logic
- Computer Architecture
- Hardware/Software Co-Design
- Embedded Systems
- Low-Level Programming

LINKS

GitHub: github.com/USERNAME

LinkedIn: linkedin.com/in/USERNAME

Portfolio: WEBSITE